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Saveetha Institute of Medical And Technical Sciences
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Decision Tree-Based Loan Approval Prediction

Using Decision Tree with Inductive Learning

Loan Applicant → Decision Tree → Approved / Rejected

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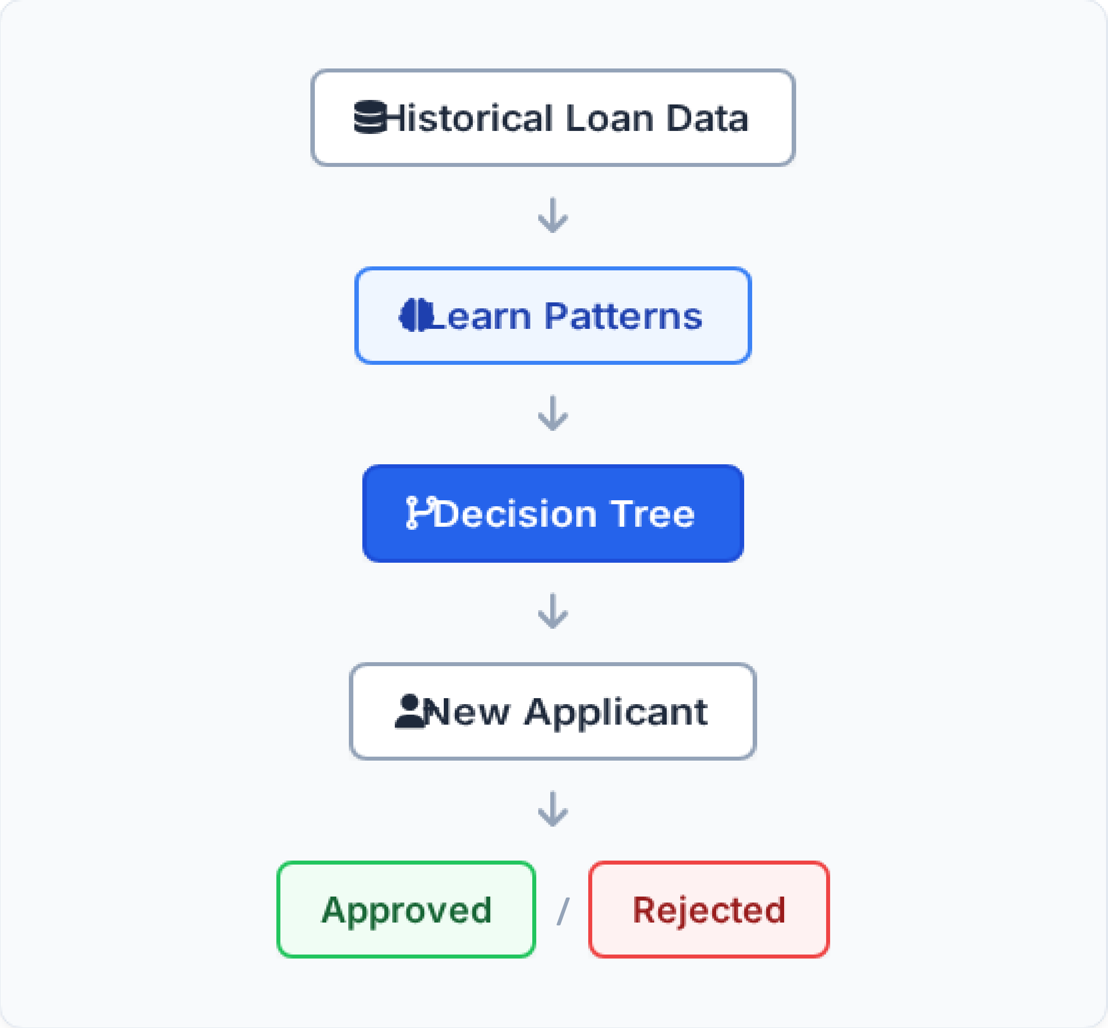
Introduction

- Loan approval depends on several applicant characteristics.
- Important factors include **income, credit history, loan amount, education,** and **property area**.
- Historical loan records contain examples of approved and rejected applications.
- **Inductive Learning** learns useful patterns from these examples.
- A **Decision Tree** represents the learned patterns as decision rules.
- The trained model predicts the loan status of a new applicant.



Problem Statement and Objectives

Problem Statement

- Manual loan evaluation can be time-consuming.
- Many applicant factors need to be considered simultaneously.
- Identifying useful patterns manually is difficult and prone to bias.
- An automated prediction model can assist with efficient initial loan screening.

Objectives

1. Collect historical loan application data.
2. Clean and preprocess the dataset.
3. Apply Inductive Learning techniques.
4. Construct a Decision Tree model.
5. Predict loan approval or rejection.
6. Evaluate model performance mathematically.

Loan Dataset and Features

Input Features

Gender	Married
Education	Applicant Income
Coapplicant Income	Loan Amount
Loan Term	Credit History
Property Area	

Target Variable

Loan Status

Approved | Rejected

Applicant Information

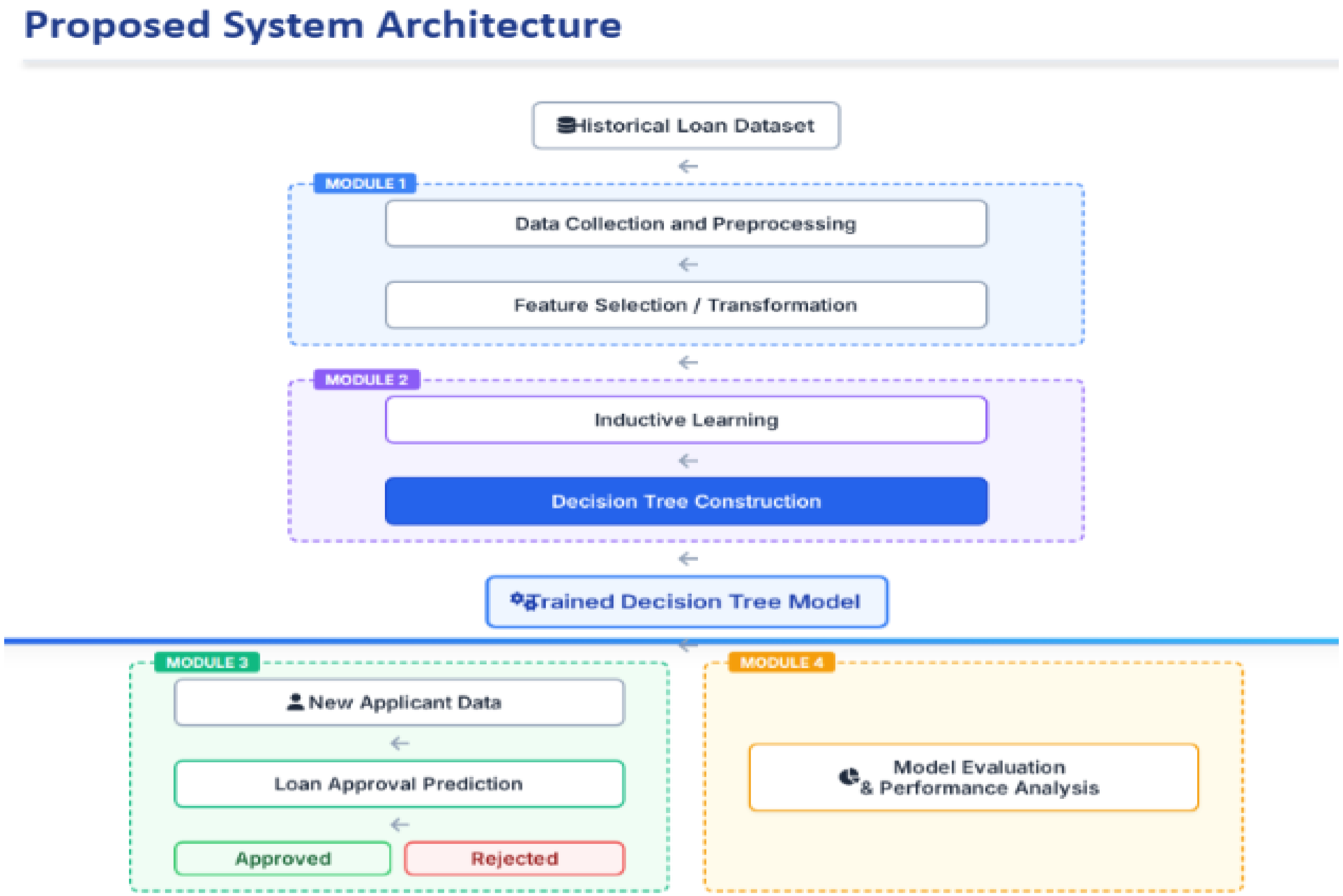
↓

Income + Credit History + Loan Amount + Education + Property Area

↓

Loan Status

Proposed System Architecture



Module 1 – Loan Data Collection and Preproces

MODULE 1

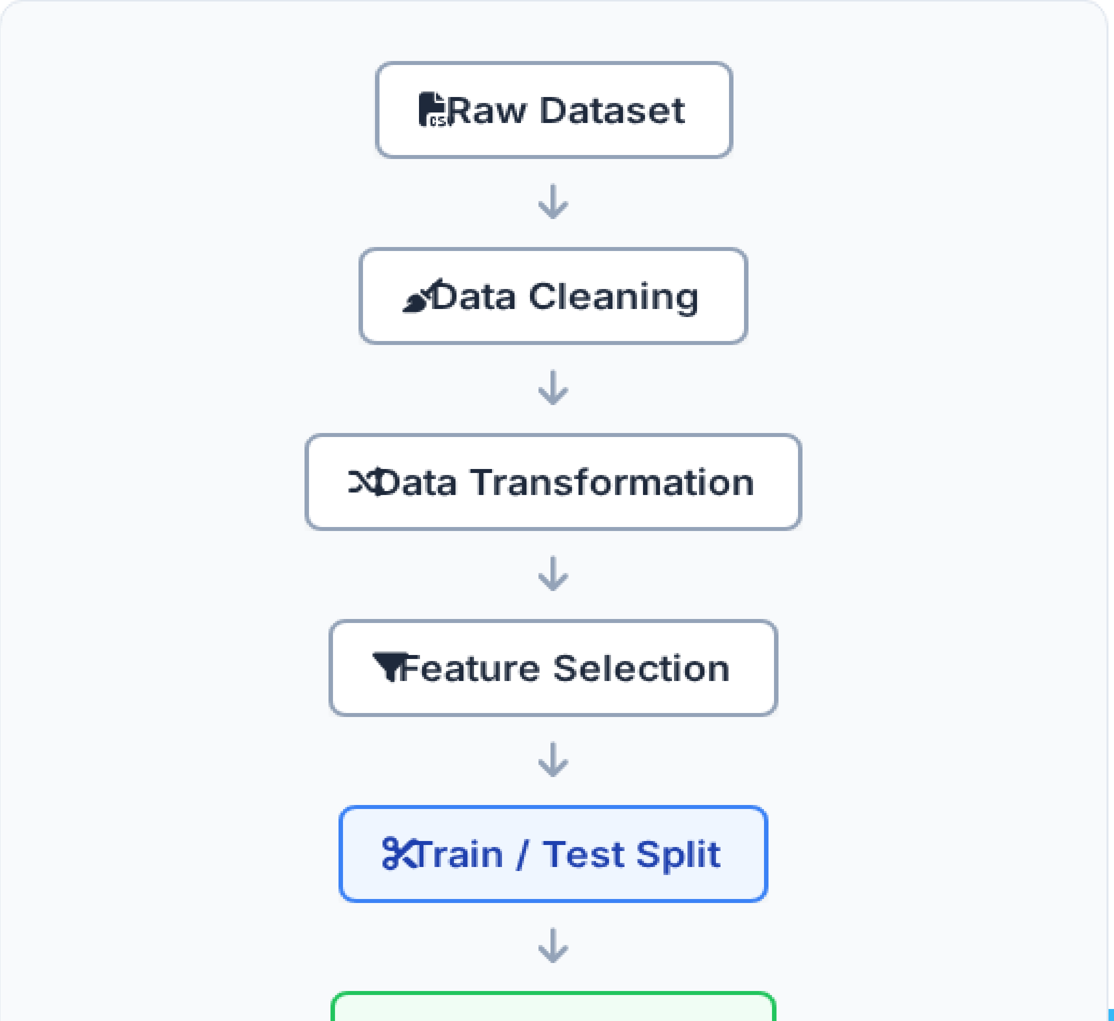
Purpose:

Prepare raw loan data for machine learning.

Steps:

- Collect historical loan records.
- Identify relevant applicant attributes.
- Handle missing values.
- Convert categorical values into numerical form.
- Select relevant features.
- Divide data into training and testing sets.

> Output: Clean and structured loan dataset.



Module 1 – Dataset Preparation

MODULE 1

Important Input Features:

Applicant Income Financial earning

Loan Amount Requested amount

Credit History Previous repayment record

Education Education level

Property Area Property location category

Loan Term Repayment period

Target: Loan Status **Approved** / **Rejected**

** Illustrative Example*

Income	Loan Amt	Credit Hist.	Property	Loan Status
$5,849	$120k	Good (1)	Urban	Approved
$4,583	$150k	Good (1)	Rural	Approved
$3,000	$100k	Bad (0)	Urban	Rejected
$2,583	$130k	Good (1)	Semiurban	Approved
$6,000	$200k	Bad (0)	Urban	Rejected

Module 2 – Inductive Learning

MODULE 2

"Inductive Learning learns general patterns from known examples and applies those patterns to new examples."

Training Examples → Known Loan Results → Identify Patterns → Build Decision Model → Predict New Applicant

Example 1:

Good Credit History + Sufficient Income **Approved Pattern**

Example 2:

Poor Credit History + Insufficient Income **Rejected Pattern**

Inductive Learning = Learning approach

Decision Tree = Algorithm/model used to represent the learned decision rules

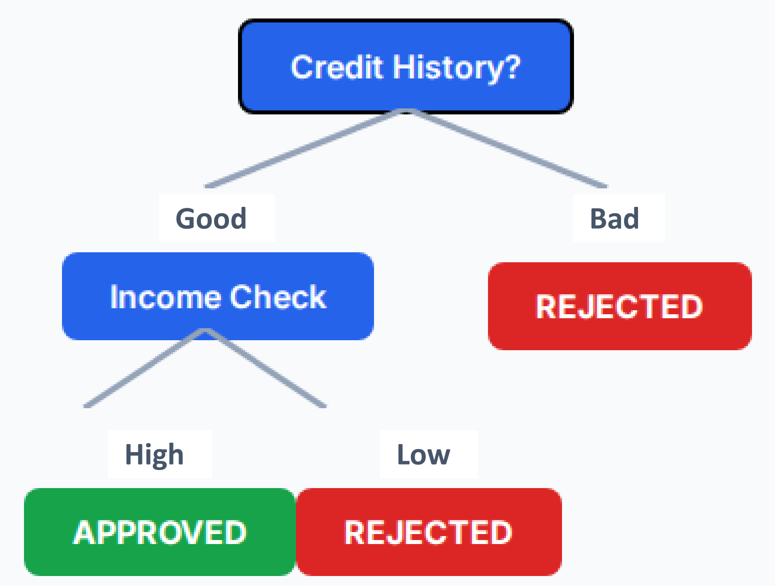
Module 2 – Decision Tree Construction

MODULE 2

Tree Components:

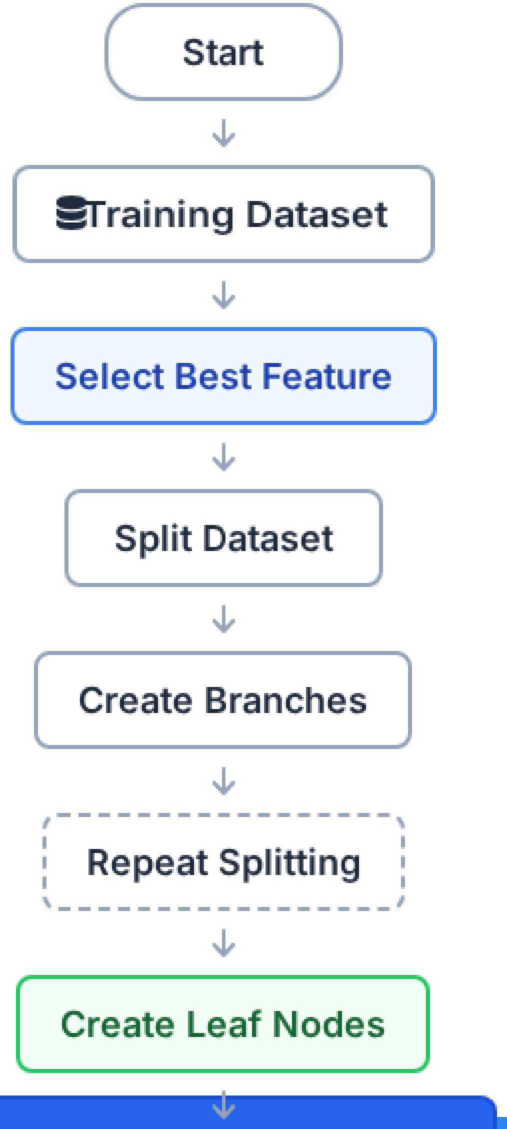
Decision Tree Construction

- The Decision Tree converts the learned patterns into **decision rules**.
- The **Root Node** represents the first and most important decision.
- **Decision Nodes** apply conditions to divide the data.
- **Branches** represent the possible outcomes of each condition.
- **Leaf Nodes** represent the final prediction: **Approved or Rejected**.
- The tree follows a sequence of decisions to classify a new applicant.



Decision Tree Learning Process

MODULE 2



Splitting Criteria:

The algorithm needs a mathematical way to decide which feature is "best" to split the data at each node.

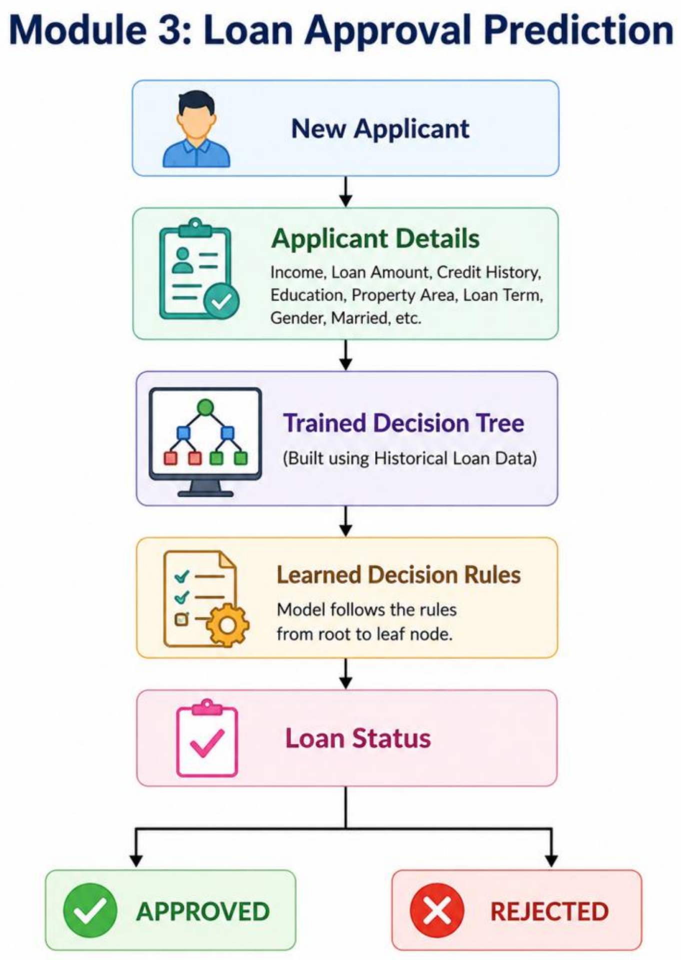
Possible metrics include:

- **Gini Index** – measures the impurity of the data.
  Information Gain – measures the reduction in uncertainty

Module 3: Loan Approval Prediction

Loan Approval Prediction

- The trained Decision Tree is used to evaluate a **new loan applicant**.
- The applicant's details are given as input to the model.
- Important inputs include **income, loan amount, credit history, education, and property area**.
- The model follows the learned decision rules from the Decision Tree.
- The applicant is classified into one of two classes.
- Final output: **Loan Approved or Loan Rejected**.



Module 3 – Loan Approval Prediction

MODULE 3

The trained Decision Tree is now used to evaluate new applicant data.

Input

- Applicant Income
- Coapplicant Income
- Loan Amount
- Credit History
- Education
- Property Area
- *Other selected features...*

New Applicant

↓

Applicant Details

↓

Trained Decision Tree

↓

Decision Rules

Output

Loan Status

APPROVED

OR

REJECTED

Module 3 – Sample Loan Prediction

MODULE 3

** Illustrative Examples*

Applicant A

Credit History: Good

Income: High

Loan Amount: Suitable

Prediction: APPROVED

Applicant B

Credit History: Bad

Income: Low

Loan Amount: High

Prediction: REJECTED

New Applicant → Decision Tree → APPROVED / REJECTED

Module 4 – Model Evaluation & Performance

MODULE 4

"The model must be tested on unseen data to determine how accurately it predicts loan status."

Evaluation Metrics:

- The trained Decision Tree is tested using **unseen test data**.
- **Accuracy** measures the overall percentage of correct predictions.
- **Precision** measures the correctness of predicted positive cases.
- **Recall** measures how well the model identifies actual positive cases.
- **F1-Score** provides a balance between precision and recall.
- A **Confusion Matrix** shows correct and incorrect predictions in detail

Confusion Matrix

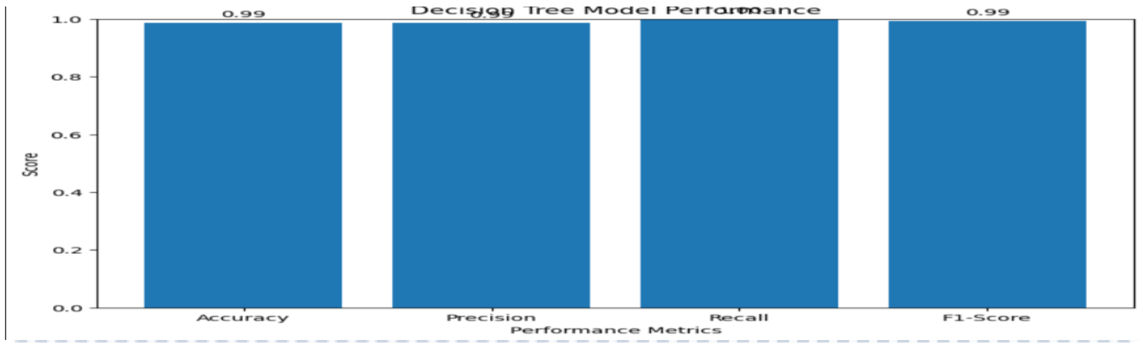
	Predicted Class
	Approved

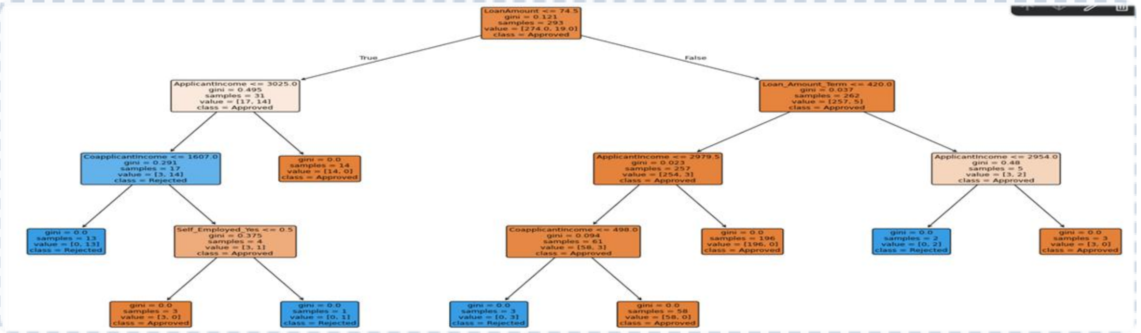
Module 4 – Results and Analysis

MODULE 4

Accuracy	Precision
98.65%	98.57%

Recall	F1-Score
100.00%	99.28%





Proposed System Architecture

Conclusion

MODULE 1

Loan data is collected, cleaned, transformed, and prepared.

MODULE 2

Inductive Learning identifies patterns and the Decision Tree represents those patterns as rules.

MODULE 3

The trained model predicts whether a new loan application is Approved or Rejected.

MODULE 4

The model is evaluated using Accuracy, Precision, Recall, and F1-Score.

FINAL PROJECT WORKFLOW

M1: Data Preparation → M2: Inductive Learning + DT → M3: Loan Prediction → M4: Model Evaluation

Thank You

Proposed System Architecture